

JEE MAIN REVISION MASTER: VECTOR ALGEBRA

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 36 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Dot Product, Projections & Orthogonality	17 PYQs	Scalar Product
High	Scalar Triple Product (Box Product) & Coplanarity	13 PYQs	Volume & Coplanarity
Moderate	Cross Product & Geometric Areas	4 PYQs	Vector Product
Moderate	Vector Triple Product & Lagrange Identity	3 PYQs	Triple Products
Standard	Linear Combination & Vector Equations	0 PYQs	Vector Equations
Standard	Section Formula, Collinearity & Centroids	0 PYQs	Geometry & Position Vectors

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Dot Product, Projections & Orthogonality	$\vec{a} \cdot \vec{b} = \vec{a} \vec{b} \cos\theta$, $\vec{a} \perp \vec{b} \iff \vec{a} \cdot \vec{b} = 0$, $\text{Proj}_{\vec{b}} \vec{a} = \frac{\vec{a} \cdot \vec{b}}{ \vec{b} ^2} \vec{b}$
Scalar Triple Product (Box Product) & Coplanarity	$[\vec{a}, \vec{b}, \vec{c}] = \vec{a} \cdot (\vec{b} \times \vec{c}) = \begin{vmatrix} a_x & a_y & a_z \\ b_x & b_y & b_z \\ c_x & c_y & c_z \end{vmatrix}$, $[\vec{a}, \vec{b}, \vec{c}] = 0 \iff \text{coplanar}$
Cross Product & Geometric Areas	$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_x & a_y & a_z \\ b_x & b_y & b_z \end{vmatrix}$, $\text{Area}_{\triangle} = \frac{1}{2} \vec{a} \times \vec{b} $, $\hat{n} = \frac{\vec{a} \times \vec{b}}{ \vec{a} \times \vec{b} }$
Vector Triple Product & Lagrange Identity	$\vec{a} \times (\vec{b} \times \vec{c}) = (\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\vec{c}$, $(\vec{a} \times \vec{b}) \times \vec{c} = (\vec{a} \cdot \vec{c})\vec{b} - (\vec{b} \cdot \vec{c})\vec{a}$
Linear Combination & Vector Equations	$\vec{r} = x\vec{a} + y\vec{b} + z(\vec{a} \times \vec{b})$, $\text{or take } \vec{a} \times (\vec{r} \times \vec{b}) \text{ to decouple}$
Section Formula, Collinearity & Centroids	$\vec{r} = \frac{m\vec{b} + n\vec{a}}{m+n}$, $\vec{G} = \frac{\vec{a} + \vec{b} + \vec{c}}{3}$, $\vec{a} = \lambda \vec{b} \iff \text{collinear}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Dot Product, Projections & Orthogonality (Very High Yield — 17 PYQs)

Core Principle: Scalar product $\vec{a} \cdot \vec{b} = |\vec{a}||\vec{b}|\cos\theta = a_x b_x + a_y b_y + a_z b_z$. Orthogonality condition: $\vec{a} \perp \vec{b} \iff \vec{a} \cdot \vec{b} = 0$. Scalar projection of $\vec{a}$ on $\vec{b}$: $\text{Proj}_{\vec{b}}(\vec{a}) = \frac{\vec{a} \cdot \vec{b}}{|\vec{b}|^2} \vec{b}$.

■ **Caution / JEE Pitfall:** 1. Vector projection of $\vec{a}$ along $\vec{b}$ is $\left(\frac{\vec{a} \cdot \vec{b}}{|\vec{b}|^2}\right)\vec{b}$. 2. $|\vec{a} + \vec{b}|^2 = |\vec{a}|^2 + |\vec{b}|^2 + 2(\vec{a} \cdot \vec{b})$ (do not write $2|\vec{a}||\vec{b}|$ unless collinear!).

■ **Speed Trick:** Unit vector along the angle bisector of $\vec{a}$ and $\vec{b}$: $\vec{u} = \frac{\vec{a}}{|\vec{a}|} + \frac{\vec{b}}{|\vec{b}|}$.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "The value of λ for which the vectors $\vec{a}=2\hat{i}+\lambda\hat{j}+k\hat{k}$ and $\vec{b}=\hat{i}+2\hat{j}+3k\hat{k}$ are orthogona..." [Ans: B]
- [2026 Shift] Q#3 (MCQ): "If $\vec{a}=2\hat{i}+2\hat{j}-k\hat{k}$, $\vec{b}=\alpha\hat{i}+\beta\hat{j}+2k\hat{k}$ and $|\vec{a}+\vec{b}|=|\vec{a}-\vec{b}|$, then $\alpha+\beta=...$ " [Ans: D]
- [2025 Shift] Q#1 (MCQ): "If $|\vec{a}|=10, |\vec{b}|=2$ and $\vec{a} \cdot \vec{b}=12$, then the value of $|\vec{a} \times \vec{b}|$ is..." [Ans: N/A]

3.2 Scalar Triple Product (Box Product) & Coplanarity (High Yield — 13 PYQs)

Core Principle: Scalar triple product $[\vec{a} \vec{b} \vec{c}] = \vec{a} \cdot (\vec{b} \times \vec{c}) = \det([a_i])$. Three vectors are coplanar if and only if $[\vec{a} \vec{b} \vec{c}] = 0$. Volume of parallelepiped $V = |[\vec{a} \vec{b} \vec{c}]|$; tetrahedron volume $V_{\text{tet}} = (1/6)|[\vec{a} \vec{b} \vec{c}]|$.

■ **Caution / JEE Pitfall:** 1. Cyclic order preserves sign: $[\vec{a} \vec{b} \vec{c}] = [\vec{b} \vec{c} \vec{a}] = [\vec{c} \vec{a} \vec{b}]$. Non-cyclic swap negates: $[\vec{a} \vec{c} \vec{b}] = -[\vec{a} \vec{b} \vec{c}]$. 2. If any two vectors are identical or collinear, the box product is strictly 0.

■ **Speed Trick:** Four position points A, B, C, D are coplanar if and only if $[\vec{AB} \vec{AC} \vec{AD}] = 0$.

Representative Source PYQs:

- [2026 Shift] Q#2 (MCQ): "If $\vec{a}=\hat{i}+\hat{j}+k\hat{k}$, $\vec{b}=\hat{j}-k\hat{k}$ and $\vec{a} \times \vec{c}=\vec{b}$, $\vec{a} \cdot \vec{c}=3$, then $\vec{c}$ i..." [Ans: C]
- [2024 Shift] Q#2 (MCQ): "If a, b and c are three non-coplanar vectors and p, q and r are vectors defined by $p=\frac{a}{[abc]}, q=\frac{b}{[abc]}, r=\frac{c}{[abc]}$..." [Ans: N/A]
- [2024 Shift] Q#3 (MCQ): "The vectors $\vec{AB}=3\hat{i}+4k\hat{k}$ and $\vec{AC}=5\hat{i}-2\hat{j}+4k\hat{k}$ are the sides of a $\triangle ABC$, The length of the median through A is..." [Ans: N/A]

3.3 Cross Product & Geometric Areas (Moderate Yield — 4 PYQs)

Core Principle: Vector cross product $\vec{a} \times \vec{b} = |\vec{a}||\vec{b}|\sin\theta \hat{n}$. Collinearity condition: $\vec{a} \parallel \vec{b}$ iff $\vec{a} \times \vec{b} = \vec{0}$. Area of triangle with sides $\vec{a}$, $\vec{b}$ is $(1/2)|\vec{a} \times \vec{b}|$; parallelogram area is $|\vec{a} \times \vec{b}|$; parallelogram with diagonals $\vec{d}_1$, $\vec{d}_2$ is $(1/2)|\vec{d}_1 \times \vec{d}_2|$.

■ **Caution / JEE Pitfall:** 1. Cross product is anti-commutative: $\vec{a} \times \vec{b} = -(\vec{b} \times \vec{a})$. 2. Lagrange's identity: $|\vec{a} \times \vec{b}|^2 = |\vec{a}|^2 |\vec{b}|^2 - (\vec{a} \cdot \vec{b})^2$.

■ **Speed Trick:** Area of a parallelogram given diagonal vectors $\vec{d}_1$, $\vec{d}_2$ is $(1/2)|\vec{d}_1 \times \vec{d}_2|$, NOT $|\vec{d}_1 \times \vec{d}_2|$!

Representative Source PYQs:

- [2022 Shift] Q#1 (MCQ): "If $\alpha = i - 3j$, $\beta = i + 2j - k$, then express β in the form $\beta = \beta_1 + \beta_2$ where $\beta_1 \dots$ " [Ans: N/A]
- [2021 Shift] Q#1 (MCQ): "If the area of the parallelogram with a and b as two adjacent sides is 15 sq units, then the area of the parallelogram having $3a + 2b$ as..." [Ans: N/A]
- [2021 Shift] Q#3 (MCQ): "The diagonals of a parallelogram are the vectors $3i + 6j - 2k$ and $-i - 2j - 8k$. Then the length of the sh..." [Ans: N/A]

3.4 Vector Triple Product & Lagrange Identity (Moderate Yield — 3 PYQs)

Core Principle: Vector triple product expands via the BAC-CAB identity: $\vec{a} \times (\vec{b} \times \vec{c}) = (\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\vec{c}$. It lies in the plane of $\vec{b}$ and $\vec{c}$, perpendicular to $\vec{a}$.

■ **Caution / JEE Pitfall:** Vector triple product is non-associative: $\vec{a} \times (\vec{b} \times \vec{c}) \neq (\vec{a} \times \vec{b}) \times \vec{c}$. Notice that $(\vec{a} \times \vec{b}) \times \vec{c} = (\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\vec{c}$.

■ **Speed Trick:** Mnemonic: 'BAC minus CAB' — dot product of outer vectors multiplied by inner vector, minus dot product of first two multiplied by the third.

Representative Source PYQs:

- [2023 Shift] Q#2 (MCQ): "The component of i in the direction of the vector $i + j + 2k$ is..." [Ans: N/A]
- [2022 Shift] Q#4 (MCQ): "If $|a \times b|^2 + |a \cdot b|^2 = 36$ and $|a| = 3$, then $|a|$ is equal to..." [Ans: D]
- [2019 Shift] Q#4 (MCQ): "A unit vector perpendicular to the plane containing the vector $i + 2j + k$ and $-2i + j + 3k$ is..." [Ans: N/A]

3.5 Linear Combination & Vector Equations (Standard Yield — 0 PYQs)

Core Principle: Solving vector equations like $\vec{r} \times \vec{a} = \vec{b}$ and $\vec{r} \cdot \vec{c} = d$. Express $\vec{r}$ as a linear basis expansion $\vec{r} = x\vec{a} + y\vec{b} + z(\vec{a} \times \vec{b})$ or take vector cross/dot products with known vectors on both sides.

■ **Caution / JEE Pitfall:** When $\vec{r} \times \vec{a} = \vec{b}$, note that $\vec{b} \perp \vec{a}$ (so $\vec{a} \cdot \vec{b} = 0$). The general solution is $\vec{r} = \lambda \vec{a} + \frac{\vec{a} \times \vec{b}}{|\vec{a}|^2}$.

■ **Speed Trick:** If $\vec{r} \times \vec{a} = \vec{b}$, taking the cross product with $\vec{a}$ gives $\vec{a} \times (\vec{r} \times \vec{a}) = \vec{a} \times \vec{b}$, decoupling $\vec{r}$ directly.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "The value of λ for which the vectors $\vec{a} = 2i + \lambda j + k$ and $\vec{b} = i + 2j + 3k$ are orthogona..." [Ans: B]
- [2026 Shift] Q#2 (MCQ): "If $\vec{a} = i + j + k$, $\vec{b} = j - k$ and $\vec{a} \times \vec{c} = \vec{b}$, $\vec{a} \cdot \vec{c} = 3$, then $\vec{c}$ i..." [Ans: C]

3.6 Section Formula, Collinearity & Centroids (Standard Yield — 0 PYQs)

Core Principle: Section formula for point dividing line segment in ratio $m:n$: $\vec{r} = \frac{m\vec{b} + n\vec{a}}{m+n}$ (internal) and $\vec{r} = \frac{m\vec{b} - n\vec{a}}{m-n}$ (external). Centroid of triangle $\vec{G} = \frac{\vec{a} + \vec{b} + \vec{c}}{3}$. Three points are collinear if $\vec{AB} = \lambda \vec{BC}$.

■ **Caution / JEE Pitfall:** Collinear vectors can be antiparallel (negative scalar λ). External division formula carries an explicit minus sign in numerator and denominator: $\frac{m\vec{b} - n\vec{a}}{m-n}$.

■ **Speed Trick:** In a tetrahedron with vertices $\vec{a}, \vec{b}, \vec{c}, \vec{d}$, the centroid is $\vec{G} = \frac{\vec{a} + \vec{b} + \vec{c} + \vec{d}}{4}$.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "The value of λ for which the vectors $\vec{a} = 2\hat{i} + \lambda\hat{j} + k\hat{k}$ and $\vec{b} = \hat{i} + 2\hat{j} + 3k\hat{k}$ are orthogona..." [Ans: B]
- [2026 Shift] Q#2 (MCQ): "If $\vec{a} = \hat{i} + \hat{j} + k\hat{k}, \vec{b} = \hat{j} - k\hat{k}$ and $\vec{a} \times \vec{c} = \vec{b}, \vec{a} \cdot \vec{c} = 3$, then $\vec{c}$ is..." [Ans: C]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

The value of λ for which the vectors $\vec{a} = 2\hat{i} + \lambda\hat{j} + k\hat{k}$ and $\vec{b} = \hat{i} + 2\hat{j} + 3k\hat{k}$ are orthogonal is

- (A) $(5/2)$
- (B) $(-5/2)$ [Correct]
- (C) $(2/5)$
- (D) $(-2/5)$

Correct Answer: B

Solution Analysis: For two vectors to be orthogonal, their dot product must be zero. Given, $\vec{a} = 2\hat{i} + \lambda\hat{j} + k\hat{k}$ and $\vec{b} = \hat{i} + 2\hat{j} + 3k\hat{k}$. Now, $\vec{a} \cdot \vec{b} = 2(1) + \lambda(2) + 1(3) = 2 + 2\lambda + 3 = 5 + 2\lambda$. Since the vectors are orthogonal, $5 + 2\lambda = 0$ $2\lambda = -5$ $\lambda = (-5/2)$ So, the correct option is Option B $(-5/2)$

Benchmark Problem 2: KCET 2026 (Q#2)

If $\vec{a} = \hat{i} + \hat{j} + k\hat{k}, \vec{b} = \hat{j} - k\hat{k}$ and $\vec{a} \times \vec{c} = \vec{b}, \vec{a} \cdot \vec{c} = 3$, then $\vec{c}$ is

- (A) $(5/3)\hat{i} - (2/3)\hat{j} + (2/3)\hat{k}$
- (B) $(5/3)\hat{i} + (2/3)\hat{j} - (2/3)\hat{k}$
- (C) $(5/3)\hat{i} + (2/3)\hat{j} + (2/3)\hat{k}$ [Correct]
- (D) $-(5/3)\hat{i} + (2/3)\hat{j} + (2/3)\hat{k}$

Correct Answer: C

Solution Analysis: Let $\vec{c} = x\hat{i} + y\hat{j} + z\hat{k}$. Given, $\vec{a} = \hat{i} + \hat{j} + k\hat{k}$. Now, $\vec{a} \times \vec{c} = \begin{vmatrix} \hat{i} & \hat{j} & k \\ 1 & 1 & 1 \\ x & y & z \end{vmatrix}$. Expanding, $\vec{a} \times \vec{c} = (z-y)\hat{i} + (x-z)\hat{j} + (y-x)k$. But it is given that $\vec{a} \times \vec{c} = \vec{b}$ and $\vec{b} = \hat{j} - k\hat{k}$. So, $(z-y)\hat{i} + (x-z)\hat{j} + (y-x)k = 0\hat{i} + 1\hat{j} - 1k$. Comparing coefficients, $z-y=0$ $x-z=1$ $y-x=-1$. From $z-y=0$, $z=y$. Also, using $x-z=1$ and $z=y$, $x-y=1$. Now given, $\vec{a} \cdot \vec{c} = 3$. So, $(\hat{i} + \hat{j} + k) \cdot (x\hat{i} + y\hat{j} + z\hat{k}) = 3$ $x + y + z = 3$. Substituting $z=y$ and $x=y+1$, $(y+1) + y + y = 3$ $3y + 1 = 3$ $3y = 2$ $y = 2/3$. Then $z = 2/3$ and $x = 5/3$. So, $\vec{c} = (5/3)\hat{i} + (2/3)\hat{j} + (2/3)\hat{k}$.

Benchmark Problem 3: KCET 2023 (Q#2)

The component of $\mathbf{i}$ in the direction of the vector $\mathbf{i} + \mathbf{j} + 2\mathbf{k}$ is

- (A) 6
- (B) $6\sqrt{6}$
- (C) $\frac{\sqrt{6}}{6}$
- (D) $\sqrt{6}$

Correct Answer: None

Solution Analysis: We know that component of $\mathbf{a}$ in the direction of $\mathbf{b}$ is $\frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{b}|}$. Let $\mathbf{b} = \mathbf{i} + \mathbf{j} + 2\mathbf{k}$ and $\mathbf{a} = \mathbf{i}$. Then, we have $|\mathbf{a}| = 1$ and $|\mathbf{b}| = \sqrt{6}$. Now, the component of $\mathbf{i}$ in the direction $\mathbf{i} + \mathbf{j} + 2\mathbf{k}$ is $\frac{1 \cdot 1}{\sqrt{6}} = \frac{1}{\sqrt{6}}$.

Benchmark Problem 4: KCET 2022 (Q#1)

If $\alpha = \mathbf{i} - 3\mathbf{j}$, $\beta = \mathbf{i} + 2\mathbf{j} - \mathbf{k}$, then express β in the form $\beta = \beta_1 + \beta_2$ where β_1 is parallel to α and β_2 is perpendicular to α , then β_1 is given by

- (A) $-\frac{1}{2}\mathbf{i} + \frac{3}{2}\mathbf{j}$
- (B) $\frac{5}{8}(\mathbf{i} + 3\mathbf{j})$
- (C) $\mathbf{i} - 3\mathbf{j}$
- (D) $\mathbf{i} + 3\mathbf{j}$

Correct Answer: None

Solution Analysis: Given $\alpha = \mathbf{i} - 3\mathbf{j}$, $\beta = \mathbf{i} + 2\mathbf{j} - \mathbf{k}$. $\beta = \beta_1 + \beta_2$. β_1 is parallel to $\alpha \Rightarrow \beta_1 = \lambda\alpha \Rightarrow \beta_1 = \lambda(\mathbf{i} - 3\mathbf{j})$. Also, $\beta = \beta_1 + \beta_2 \Rightarrow \beta_2 = \beta - \beta_1 = (\mathbf{i} + 2\mathbf{j} - \mathbf{k}) - \lambda(\mathbf{i} - 3\mathbf{j}) = (1 - \lambda)\mathbf{i} + (2 + 3\lambda)\mathbf{j} - \mathbf{k}$. It is given β_2 is perpendicular to α . $\therefore (1 - \lambda)(1) + (2 + 3\lambda)(-3) + (-1)(0) = 0 \Rightarrow 1 - \lambda - 6 - 9\lambda = 0 \Rightarrow -5 - 8\lambda = 0 \Rightarrow \lambda = -\frac{5}{8}$. $\therefore \beta_1 = -\frac{5}{8}(\mathbf{i} - 3\mathbf{j}) = \frac{5}{8}(-\mathbf{i} + 3\mathbf{j})$.